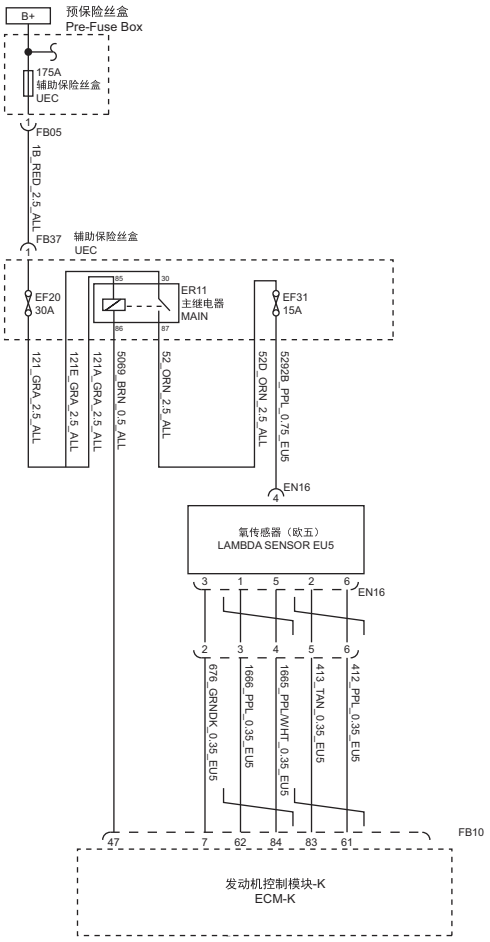
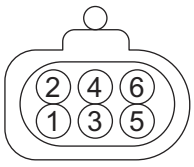
P0130、P015A、P064D、P2195、P2196、P2243、P2626

Description of fault diagnosis codes
P0130-11：short to ground at IA,IP, UN, VG of lambda sensor
P0130-12：short to battery at IA,IP, UN, VG of lambda sensor
P0030-84：LSU sensor calculated temperature is below the minimum limit
P0130-85：LSU sensor calculated temperature exceeds the maximum limit
P0130-F0：The analog signal for O2 ratio from sensor after PT1, during calibration mode exceeds maximum threshold value
P0130-F1：The analog signal for O2 ratio from sensor after PT1, during calibration mode is below minimum threshold value
P015A-00：Fault to indicate Dynamics of the sensor signal too small
P064D-29：Fault check to indicate SPI chip error of lambda sensor
P2195-64：Oxygen concentration implausibly high (Overrun detection)
P2196-64：Oxygen concentration implausibly low (Overrun detection)
P2196-85：O2 value above the max threshold
P2243-11：low battery voltage at the SPI chip of lambda sensor
P2626-29：Open circuit at the lambda sensor Virtual ground pin
P2626-2A：Open circuit at the lambda sensor Nernst cell pin
P2626-F0：open circuit at the lambda sensor pump current pin - IP (lambda = 1 detection)
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Oxygen sensor fault • Exhaust manifold leakage • Engine control module fault.
Description of circuit diagram
The heated oxygen sensor is used to monitor the fuel control. The engine control module provides 5v reference voltage circuit to the oxygen sensor, and also provides grounding to low reference voltage circuit. The signal voltage provided by oxygen sensor will vary with the oxygen concentration of exhaust manifold.

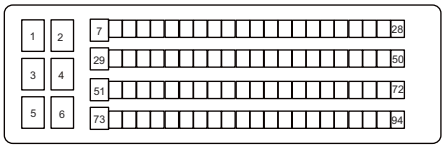
Circuit diagram



EN16

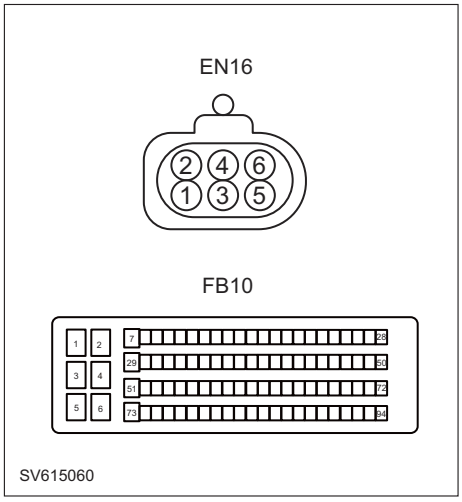


FB10



SV615022

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse EF31 for damage. 2 Check battery capacity. 3 Check the harness plugs of engine control module and oxygen sensor for damage, poor contact, aging and looseness. 4 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check the exhaust leakage.	
		1 Start up the engine to perform idle running. 2 Check the exhaust manifold (front end of oxygen sensor) for leakage. 3 Check whether the result is normal?
		No → Repair and replace the exhaust manifold.
Yes ↓		
Step 3	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Are there any other fault codes except for P0130, P015A, P064D, P2195, P2196, P2243 and P2626?
		Yes → Refer to: DTC Summary List (EU5).
No ↓		

Step 4	Check the circuit of oxygen sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of oxygen sensor. 3 Disconnect the harness plug of engine control module. 4 Measure the resistance between the harness plug of oxygen sensor and that of engine control module. Measure circuit: the resistance between EN16 plug terminal 1 and FB10 plug terminal 62. Measure circuit: the resistance between EN16 plug terminal 2 and FB10 plug terminal 83. Measure circuit: the resistance between EN16 plug terminal 5 and FB10 plug terminal 84. Measure circuit: the resistance between EN16 plug terminal 6 and FB10 plug terminal 61. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the circuit of the oxygen sensor(short circuit).
	1 Check the resistance between the harness plug of oxygen sensor and ground. Measure circuit: The resistance between terminal 1 of EN16 plug and ground. Measure circuit: The resistance between terminal 2 of EN16 plug and ground. Measure circuit: The resistance between terminal 5 of EN16 plug and ground. Measure circuit: The resistance between terminal 6 of EN16 plug and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of oxygen sensor and ground. Measure circuit: The voltage between terminal 1 of EN16 plug and ground. Measure circuit: The voltage between terminal 2 of EN16 plug and ground. Measure circuit: The voltage between terminal 5 of EN16 plug and ground. Measure circuit: The voltage between terminal 6 of EN16 plug and ground. **Standard value: 0V** 3 Check whether the result is normal?

	No → Repair or replace the faulted parts.
Yes ↓	
Step 6	Check the signal power of oxygen sensor.
	1 Connect the harness plug of engine control module. 2 Turn the ignition switch to ON position. 3 Measure the voltage between oxygen sensor hardness and ground. Measure circuit: The voltage between terminal 1 of EN16 plug and ground. **Standard value: 4.8~5.2V** 4 Check whether the result is normal?
	No → Replace engine control module.
Yes ↓	
Step 7	Check the signal of oxygen sensor.
	1 Turn the ignition switch to OFF. 2 Connect the harness plug of oxygen sensor. 3 Turn the ignition switch to ON position. 4 Measure the voltage between oxygen sensor and ground. Measure circuit: The voltage between terminal 6 of EN16 plug and ground. **Standard value: 2~3V** 5 Check whether the result is normal?
	No → Replace the oxygen sensor.
Next ↓	
Step 8	Complete the check and confirm whether the system works normally nor not.